

Integral Secant Distributions and Improved Bounds for Complete (k, n) -Arcs

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Abstract

We study complete higher arcs in finite projective planes through the numbers of maximal secants incident with each point. Splitting the exact block-pair count between the arc and its complement gives sharp integer bounds for both degree sequences. Their real relaxation is the classical incidence bound, but the integer bounds retain a linear obstruction at every rational equality parameter. For each factorization $\lambda = uv$, this gives an explicit lower bound for (k, n) -arcs for which every external point lies on at least λ n -secants, with $q = (u + v + 1)m$ and $n = (u + 1)m + 1$, improving the spectral coefficient by a positive multiple of q . In the ordinary-completeness branch over $\text{PG}(2, 3^h)$, modular stability and the maximal-secant condition strengthen the bound to

$$k \geq \frac{q^2}{3} + \frac{4q}{3} - o(q), \quad s = \frac{2q}{3} + 1.$$

An analogous parity argument over characteristic two gives a further linear improvement for double coverage. The degree-envelope theorem is stated for selected blocks of an arbitrary symmetric design; complete higher arcs, minimal multiple blocking sets, and robustly nonextendible projective codes are its geometric forms.

Keywords. complete higher arc; maximal secant; multiple blocking set; symmetric design; projective code; incidence bound; modular stability.

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1 Introduction

A (k, n) -arc in a projective plane is a set of k points meeting every line in at most n points, with some line meeting it in exactly n points. It is complete if it is maximal under inclusion, or equivalently if every external point lies on an n -secant. The minimum size of a complete (k, n) -arc is denoted by $t_n(2, q)$ in $\text{PG}(2, q)$. Lower bounds for this quantity are usually obtained by counting maximal secants or, dually, by bounding minimal multiple blocking sets; see Alabdullah–Hirschfeld [1] and Bishnoi–Mattheus–Schillewaert [2]. Recent work has also developed algebraic and curve-based methods for proving completeness itself; see, for example, Korchmáros–Nagy–Szőnyi [4].

This paper keeps the maximal secants as a family rather than eliminating their number at the first counting step. The degrees of that family on the arc and on its complement have fixed sums, and their pair counts add to the number of pairs of secants. Integer convexity gives the exact range of each pair count. The two ranges must overlap. Replacing the integer extrema by quadratic real averages recovers the classical incidence-mixing inequality; retaining them gives the new bounds.

The strongest application concerns ordinary completeness in characteristic three.

Theorem 1.1 (Characteristic-three bound). *For every $\varepsilon > 0$, there is h_0 such that, for $h \geq h_0$, every complete $(k, 2q/3 + 1)$ -arc in $\text{PG}(2, q)$, where $q = 3^h$, satisfies*

$$k \geq \frac{q^2}{3} + \left(\frac{4}{3} - \varepsilon\right)q. \quad (1.1)$$

Equivalently,

$$t_{2q/3+1}(2, q) \geq \frac{q^2}{3} + \frac{4q}{3} - o(q) \quad (q = 3^h \rightarrow \infty).$$

The incidence-mixing bound gives only $q^2/3 + q + O(1)$ in this regime. The integer degree bounds first raise the linear coefficient to $6/5$; a modular repair theorem of Szőnyi–Weiner [9] then raises it to $4/3$. The second step uses more than modular stability: each point changed in the dual repaired multiset becomes a primal line on which completeness forces a linear amount of external secant excess.

The arithmetic source of the first improvement is visible whenever the real incidence bound is tight at a rational parameter. Fix a coverage multiplicity λ , factor it as $\lambda = uv$, and put $d = u + v + 1$. For $h \in \mathbb{Z}$, define

$$c_{\text{bal}} = \frac{u(uv^2 + 4uv + 2u + 2v^2 + 4v + 1)}{2uv + u + v}, \quad (1.2)$$

$$h_* = -\frac{u(v+1)(u^2v + u^2 - v)}{2uv + u + v}, \quad (1.3)$$

$$\mu = \frac{2v}{(u+v)(2uv + u + v + 1)},$$

$$L(h) = d - \frac{h}{u}, \quad C(h) = c_{\text{bal}} + \mu(h - h_*),$$

$$c_{\mathbb{Z}} = \min_{h \in \mathbb{Z}} \max\{L(h), C(h)\}. \quad (1.4)$$

Theorem 1.2 (Rational equality families). *Let $u, v \geq 1$, $\lambda = uv$, and $d = u + v + 1$. Suppose that a $(k, (u+1)m + 1)$ -arc in a projective plane of order $q = dm$ has at least λ n -secants through every external point. Then*

$$k \geq udm^2 + c_{\mathbb{Z}}m - O_{u,v}(1). \quad (1.5)$$

Moreover, the classical incidence-mixing bound has linear coefficient

$$c_{\text{mix}} = \frac{u(uv + 3u + 2v + 3)}{2u + 1},$$

and

$$c_{\text{bal}} - c_{\text{mix}} = \frac{u(v+1)(u^2 + u + 1)}{(2u+1)(2uv + u + v)} > 0. \quad (1.6)$$

Every rational equality parameter of the continuous bound arises from exactly one ordered factorization $\lambda = uv$.

Thus the strict improvement is not an isolated rounding effect. At fixed coverage multiplicity there is one equality family for each divisor, and every family has a positive linear integrality penalty. The two integers adjacent to h_* determine $c_{\mathbb{Z}}$, so the coefficient is explicit.

The same integer incidence condition holds in arbitrary symmetric designs. In a projective plane there are two further bounds. At most $\lfloor (k-1)/(n-1) \rfloor$ n -secants pass through a point of the arc, and at most $\lfloor k/n \rfloor$ pass through an external point. These geometric bounds sharpen the general condition for complete arcs.

In characteristic two the first family in which the two first-order bounds are simultaneously tight yields a second application.

Theorem 1.3 (Characteristic-two parity bound). *For every $\varepsilon > 0$, there is h_0 such that, for $h \geq h_0$, if $q = 2^h$, $8 \mid q$, and every external point of a $(k, 3q/4 + 1)$ -arc in $\text{PG}(2, q)$ lies on at least two $(3q/4 + 1)$ -secants, then*

$$k \geq \frac{q^2}{2} + \left(\frac{73}{48} - \varepsilon \right) q. \quad (1.7)$$

The proof separates odd and even repair support. The parity of the selected secant family fixes the parity of the support, while the symmetric difference of its generator lines supplies respectively $q/4 - O(1)$ or $q/2 - O(1)$ external points of excess degree.

The projective-code translation is immediate. A point set in $\text{PG}(r-1, q)$ gives a projective $[k, r, k-s]_q$ code; maximal hyperplanes are minimum-weight codewords, and a candidate extension column is obstructed by each maximal hyperplane through it. The same integer degree theorem therefore bounds robust nonextendibility. We keep the finite-geometry language primary and record this translation after the planar applications.

The paper is organized as follows. Section 2 proves the symmetric-design inequality and identifies incidence mixing as its real relaxation. The next two sections specialize to maximal secants and prove Theorem 1.2. The modular-lift section proves Theorems 1.1 and 1.3. The final section gives the blocking-set and coding translations and records the exact computational checks used to guard the finite algebra.

2 Sharp integer degree bounds

For integers $N \geq 1$ and $R \geq 0$, write $R = aN + w$ with $0 \leq w < N$, and define

$$\Phi_{\min}(N, R) = (N - w) \binom{a}{2} + w \binom{a+1}{2}. \quad (2.1)$$

If $L \leq U$, $NL \leq R \leq NU$, put $E = R - NL = t(U - L) + w$, where $0 \leq w < U - L$, and define

$$\Phi_{\max}(N, L, U, R) = N \binom{L}{2} + LE + t \binom{U-L}{2} + \binom{w}{2}. \quad (2.2)$$

When $L = U$, the latter expression has its evident constant interpretation.

Lemma 2.1 (Balanced and filled degree sequences). *Among integer sequences $z_1, \dots, z_N$ with sum R , the minimum of $\sum_i \binom{z_i}{2}$ is $\Phi_{\min}(N, R)$. With the additional bounds $L \leq z_i \leq U$, its maximum is $\Phi_{\max}(N, L, U, R)$.*

Proof. If $z_i \geq z_j + 2$, replacing (z_i, z_j) by $(z_i - 1, z_j + 1)$ lowers the sum by $z_i - z_j - 1$. Repetition leaves only the adjacent values $a, a+1$, proving (2.1). For the maximum, the reverse exchange moves one unit from a smaller nonminimal coordinate to a larger nonmaximal one and raises the sum. Hence all but at most one coordinate lie at L or U ; counting the filled coordinates gives (2.2). $\square$

Theorem 2.2 (Exact integer incidence condition). *Let $\mathcal{D}$ be a symmetric 2 -(v, K, Λ) design whose point set is partitioned into $A \sqcup B$. Let $\mathcal{S}$ be a family of T blocks, each meeting A in s points. Suppose the selected-block degrees satisfy integer bounds*

$$L_x \leq z_x \leq U_x \quad (x \in A \sqcup B).$$

At the fixed totals

$$\sum_{x \in A} z_x = sT, \quad \sum_{x \in B} z_x = (K - s)T,$$

let $[P_A^-, P_A^+]$ and $[P_B^-, P_B^+]$ be the exact minimum and maximum of the two sums $\sum \binom{z_x}{2}$ over their respective integer boxes. Then

$$[P_B^-, P_B^+] \cap \left[\Lambda \binom{T}{2} - P_A^+, \Lambda \binom{T}{2} - P_A^- \right] \neq \emptyset. \quad (2.3)$$

For uniform boxes the endpoints are given by Lemma 2.1; heterogeneous endpoints are obtained by integer water filling.

Proof. Count triples (x, S_1, S_2) with $S_1, S_2 \in \mathcal{S}$, $S_1 \neq S_2$, and $x \in S_1 \cap S_2$. Two blocks of a symmetric design meet in exactly Λ points, so

$$\sum_{x \in A} \binom{z_x}{2} + \sum_{x \in B} \binom{z_x}{2} = \Lambda \binom{T}{2}. \quad (2.4)$$

Each sum lies in its exact integer interval. Reflecting the interval for A through the right side of (2.4) gives (2.3). $\square$

The real relaxation is familiar, but the following formulation makes the loss of information exact.

Corollary 2.3 (Incidence mixing as the real relaxation). *In the setting of Theorem 2.2,*

$$T \left(\frac{s^2}{|A|} + \frac{(K-s)^2}{|B|} - \Lambda \right) \leq K - \Lambda. \quad (2.5)$$

This is algebraically identical to the incidence-graph mixing inequality for the point set B and block family $\mathcal{S}$. Equality requires constant selected-block degree on both point classes.

Proof. Cauchy–Schwarz and (2.4) give

$$\frac{s^2 T^2}{|A|} - sT + \frac{(K-s)^2 T^2}{|B|} - (K-s)T \leq \Lambda T(T-1).$$

Divide by T and use $K = s + (K-s)$. Conversely, expanding the standard mixing inequality and substituting $K(K-1) = \Lambda(v-1)$ gives the same expression. Equality in the two Cauchy inequalities is equivalent to constant degrees. $\square$

For uniform lower bounds without an upper cap, the balancing remainder has the closed form

$$\Phi_{\min}(N, R) = \frac{R^2/N - R}{2} + \frac{w(N-w)}{2N}, \quad R \equiv w \pmod{N}, \quad 0 \leq w < N. \quad (2.6)$$

The last term is precisely what the real relaxation discards. At a resonant parameter sequence, it has linear size and cannot be absorbed into lower-order error.

3 The distribution of n -secants

Let $\mathcal{K}$ be a (k, n) -arc in a projective plane of order q . We use the standard secant notation

$$\tau_i = \#\{i\text{-secants of } \mathcal{K}\},$$

and write $\rho_i(P)$ for the number of i -secants through $P \in \mathcal{K}$, and $\sigma_i(Q)$ for the number through $Q \notin \mathcal{K}$. Only $i = n$ will occur below. Put

$$b = q^2 + q + 1 - k, \quad \tau = q + 1 - n, \quad T = \tau_n.$$

The geometry supplies two bounds that need not hold for a selected block family in an arbitrary design.

Lemma 3.1 (Secants in a pencil). *For every $P \in \mathcal{K}$ and $Q \notin \mathcal{K}$,*

$$\rho_n(P) \leq D := \left\lfloor \frac{k-1}{n-1} \right\rfloor, \quad \sigma_n(Q) \leq M := \left\lfloor \frac{k}{n} \right\rfloor. \quad (3.1)$$

Moreover,

$$\sum_{P \in \mathcal{K}} \rho_n(P) = nT, \quad \sum_{Q \notin \mathcal{K}} \sigma_n(Q) = \tau T, \quad (3.2)$$

$$\sum_{P \in \mathcal{K}} \binom{\rho_n(P)}{2} + \sum_{Q \notin \mathcal{K}} \binom{\sigma_n(Q)}{2} = \binom{T}{2}. \quad (3.3)$$

Proof. The n -secants through P have disjoint sets of $n-1$ further points of $\mathcal{K}$. Those through Q have disjoint n -point intersections with $\mathcal{K}$. This proves (3.1). The first two equations count incidences. Every two distinct projective lines meet once, either in $\mathcal{K}$ or outside it, which gives (3.3). $\square$

We will sometimes impose the stronger hypothesis

$$\sigma_n(Q) \geq \lambda \quad (Q \notin \mathcal{K}), \quad (3.4)$$

where $\lambda \geq 1$. For $\lambda = 1$, this is precisely completeness. Rather than introduce another name for (3.4), we state the multiplicity explicitly in each result.

Corollary 3.2 (Integer feasibility for the n -secants). *Assume (3.4). Then there is an integer T satisfying*

$$\left\lceil \frac{\lambda b}{\tau} \right\rceil \leq T \leq \left\lfloor \frac{k(k-1)}{n(n-1)} \right\rfloor, \quad (3.5)$$

$$[\Phi_{\min}(b, \tau T), \Phi_{\max}(b, \lambda, M, \tau T)] \cap \left[\binom{T}{2} - \Phi_{\max}(k, 0, D, nT), \binom{T}{2} - \Phi_{\min}(k, nT) \right] \neq \emptyset. \quad (3.6)$$

Proof. Equation (3.4) and (3.2) give the lower bound for T . Each n -secant contains $\binom{n}{2}$ pairs of points of $\mathcal{K}$, and each pair lies on one line, giving the upper bound. Apply Theorem 2.2 to the point–line design of the plane, using Lemma 3.1 for the degree bounds. $\square$

If

$$v_{\mathcal{K}} \equiv nT \pmod{k}, \quad 0 \leq v_{\mathcal{K}} < k, \quad v_{\text{ext}} \equiv \tau T \pmod{b}, \quad 0 \leq v_{\text{ext}} < b,$$

then the two minimum terms in (3.6) imply

$$T \left(\frac{n^2}{k} + \frac{\tau^2}{b} - 1 \right) + \frac{v_{\mathcal{K}}(k - v_{\mathcal{K}})}{kT} + \frac{v_{\text{ext}}(b - v_{\text{ext}})}{bT} \leq q. \quad (3.7)$$

Discarding the two nonnegative residue terms gives the familiar real incidence bound. Since $T \geq \lambda b/\tau$, we obtain

$$\frac{\lambda b}{\tau} \left(\frac{n^2}{k} + \frac{\tau^2}{b} - 1 \right) \leq q. \quad (3.8)$$

After denominators are cleared, (3.8) is the quadratic in Bishnoi–Mattheus–Schillewaert [2, Theorem 8.1]. Thus the classical bound is the real relaxation of (3.6); the residue terms record part of the discarded integer information.

Remark 3.3. For a complete (k, n) -arc one takes $\lambda = 1$. The stronger values of λ below mean only that every external point is on at least λn -secants. This is the same multiplicity hypothesis used in the dual form of Bishnoi–Mattheus–Schillewaert [2, Theorem 8.1].

4 Rational equality parameters

We first determine when the real incidence bound can be tight while the limiting number of n -secants through a point of the arc is an integer. Suppose that

$$\frac{n}{q} \longrightarrow \alpha, \quad \frac{k}{q^2} \longrightarrow \beta,$$

that equality holds asymptotically in (3.8), and that the limiting value of $\rho_n(P)$ is an integer $a > \lambda$. Equality in the incidence and coverage bounds gives

$$a = \frac{\lambda\alpha(1-\beta)}{(1-\alpha)\beta}, \quad \lambda(\alpha-\beta)^2 = \beta(1-\alpha). \quad (4.1)$$

If $d = a - \lambda$, eliminating β gives

$$d^2\alpha^2 - d(d+1)\alpha + a = 0, \quad (d+1)^2 - 4a = (a - \lambda - 1)^2 - 4\lambda. \quad (4.2)$$

Lemma 4.1 (Classification of the rational parameters). *The rational solutions of (4.1) are in bijection with ordered factorizations $\lambda = uv$, where $u, v \geq 1$. They are*

$$d = u + v + 1, \quad \alpha = \frac{u+1}{u+v+1}, \quad \beta = \frac{u}{u+v+1}, \quad a = (u+1)(v+1). \quad (4.3)$$

Proof. The last expression in (4.2) must be a square. Factoring the difference of squares gives two positive factors of 4λ with the same parity. Both factors are even; write them as $2u$ and $2v$. Then $uv = \lambda$, and substitution gives (4.3). Conversely, the displayed values satisfy (4.1). Their order distinguishes the two roots, so the correspondence is one-to-one. $\square$

We next control $T = \tau_n$ near one of these parameters. The argument is stated separately because it is also used in the modular refinements.

Lemma 4.2 (Number of n -secants near equality). *Fix $u, v \geq 1$, put $d = u + v + 1$, and suppose*

$$q = dm, \quad n = (u+1)m + 1, \quad k = udm^2 + cm + O(1),$$

where c remains bounded. If every external point is on at least uv n -secants, then the minimum, over the permitted values of T , of

$$F(T) = \Phi_{\min}(k, nT) + \Phi_{\min}(q^2 + q + 1 - k, vmT) - \binom{T}{2} \quad (4.4)$$

occurs at

$$T = ud(v+1)m + O_{u,v,c}(1). \quad (4.5)$$

Proof. Set $b = q^2 + q + 1 - k$. The external incidence count gives

$$T \geq T_0 := \left\lceil \frac{uvb}{vm} \right\rceil = ud(v+1)m + O(1). \quad (4.6)$$

From (2.1),

$$F(T+1) - F(T) = \sum_{i=0}^{n-1} \left\lfloor \frac{nT+i}{k} \right\rfloor + \sum_{i=0}^{vm-1} \left\lfloor \frac{vmT+i}{b} \right\rfloor - T. \quad (4.7)$$

From T_0 through an interval of length $vm + O(1)$, the first family of floors is at least $(u+1)(v+1) - 1$, and the second is at least uv . Thus (4.7) is at least $vm - (T - T_0) + O(1)$. Beyond this interval, use $\lfloor x \rfloor \geq x - 1$ and

$$\frac{n^2}{k} + \frac{v^2 m^2}{b} - 1 = \frac{1}{u(v+1)} + O(m^{-1}).$$

At the end of the first interval this gives a positive quantity of order m , and the associated real quadratic is increasing throughout the remaining permitted range. Hence a minimizer differs from T_0 by only a bounded amount. $\square$

Proof of Theorem 1.2. Lemma 4.1 proves the classification. The lower bound is automatic if $(k - udm^2)/m$ is unbounded, so pass to a subsequence on which

$$k = udm^2 + cm + O(1), \quad T = ud(v+1)m + h$$

with c bounded and $h \in \mathbb{Z}$ fixed. The second passage follows from Lemma 4.2. Expanding (4.6) to first order gives

$$c \geq L(h) + o(1) = d - \frac{h}{u} + o(1). \quad (4.8)$$

At these parameters the balanced internal degrees are $(u+1)(v+1) - 1$ and $(u+1)(v+1)$; the balanced external degrees are uv and $uv + 1$. Substitution in (2.1) gives

$$\frac{F(T)}{m} = \frac{(u+v)(2uv + u + v + 1)}{2} (C(h) - c) + o(1). \quad (4.9)$$

The integer condition (3.6) implies $F(T) \leq 0$, so $c \geq C(h) + o(1)$. The affine functions L and C are decreasing and increasing, respectively, and meet at h_* . It follows that their maximum over the integers is minimized at $\lfloor h_* \rfloor$ or $\lceil h_* \rceil$. This proves (1.5).

Expanding the real bound (3.8) gives c_{mix} . Subtraction from c_{bal} gives the factorization (1.6). $\square$

For the first three ordered factorizations, the integer coefficients are

$$(u, v) = (1, 1) : \frac{18}{5}, \quad (u, v) = (1, 2) : \frac{9}{2}, \quad (u, v) = (2, 1) : 6. \quad (4.10)$$

The first is the ordinary-completeness family $q = 3m$, $n = 2m + 1$. The other two have two n -secants through every external point and limiting ratios $n/q = 1/2$ and $3/4$.

There is also an elementary description of the parameters for which the coverage and pair-count inequalities are simultaneously tight to first order.

Proposition 4.3 (Simultaneous first-order equality). *Put $M_0 = 2uv + u + v$. Then h_* is an integer if and only if*

$$M_0 \mid u^2(u+1)(u^2 + u + 1). \quad (4.11)$$

In particular, the case $v = u$ has simultaneous first-order equality for every even u .

Proof. Polynomial division in v , followed by multiplication by $(2u+1)^2$, gives

$$(2u+1)^2 u(v+1)(u^2 v + u^2 - v) = M_0 Q(u, v) + u^2(u+1)(u^2 + u + 1)$$

for an integer polynomial Q . Since $\gcd(M_0, 2u+1) = \gcd(M_0, u) = 1$, integrality of h_* is equivalent to (4.11). When $v = u$,

$$h_* = -\frac{u(u^2 + u - 1)}{2},$$

which is integral for even u . $\square$

5 Modular stability and completeness

Assume now that the plane is $\text{PG}(2, q)$, where $q = p^e$. Dualizing the set of n -secants gives a point set $\mathcal{L}^*$ in the dual plane. The intersection of $\mathcal{L}^*$ with the line dual to a point X has size $\rho_n(X)$ or $\sigma_n(X)$, according as $X \in \mathcal{K}$ or $X \notin \mathcal{K}$. At a rational equality parameter from Section 4, the two limiting intersection numbers are

$$\lambda = uv, \quad a = (u+1)(v+1), \quad a - \lambda = u + v + 1.$$

When $u + v + 1$ is a power of p , these two numbers are congruent modulo p .

Szönyi and Weiner prove that a multiset in $\text{PG}(2, p^e)$ for which only a small number of lines fail to have one fixed intersection residue can be changed at few points into a multiset having that residue on every line [9, Theorem 1.2]. We use their theorem in the range where the number of exceptional lines is $O(q)$, so the number of changed support points is bounded while q tends to infinity.

The following observation is where completeness re-enters the modular argument.

Lemma 5.1 (Contribution of the repair support). *Fix an ordered factorization $\lambda = uv$ and suppose $d = u + v + 1$ is a power of the characteristic. Let*

$$q = dm, \quad n = (u+1)m + 1, \quad k = udm^2 + cm + o(m), \quad T = ud(v+1)m + h + o(1),$$

where $h \in \mathbb{Z}$. Suppose every external point is on at least λ n -secants, and suppose $\mathcal{L}^*$ can be changed at r distinct support points into a multiset meeting every line in $\lambda \pmod{p}$ points. If r is fixed and nonzero, then

$$c \geq L(h) + \frac{r}{u}. \quad (5.1)$$

Proof. Dualize the r support points to r lines of the original plane. Away from their pairwise intersections, a point on one of these lines has n -secant multiplicity different from $\lambda \pmod{p}$. If it is external, its multiplicity is at least $\lambda + 1$. A support line contains at most n points of $\mathcal{K}$, and the union loses only $O(r^2)$ points at pairwise intersections. Hence the total external excess over multiplicity λ is at least

$$r(q + 1 - n) - O(r^2) = rvm - O(r^2). \quad (5.2)$$

On the other hand, the incidence sum gives the exact first-order expression

$$(q + 1 - n)T - \lambda(q^2 + q + 1 - k) = uv(c - L(h))m + o(m). \quad (5.3)$$

Comparing (5.2) and (5.3) proves (5.1). $\square$

There is a useful restriction when $r = 1$. Summing the line intersections of an exact $\lambda \pmod{p}$ multiset over a pencil shows that its total multiplicity is $\lambda \pmod{p}$. Along a characteristic-compatible family the leading term of T is divisible by p , so the correction has total multiplicity $h - \lambda \pmod{p}$. Consequently, if

$$h \equiv \lambda \pmod{p}, \quad (5.4)$$

then a nonzero correction cannot have support one.

We apply this first to ordinary completeness. Write $q = 3m$, $n = 2m + 1$, and suppose

$$k = 3m^2 + cm + o(m), \quad T = 6m + h + o(1). \quad (5.5)$$

The coverage and pair-count inequalities from Sections 3 and 4 become

$$c + h \geq 3, \quad 5c - h \geq 18. \quad (5.6)$$

Lemma 5.2 (The uncorrected case is impossible). *Under (5.5), if $c < 4$ and every line meets $\mathcal{L}^*$ in $1 \pmod{3}$ points, then the exact pair identity (3.3) cannot hold.*

Proof. Near this equality parameter the balanced internal multiplicities are 3, 4, and the balanced external multiplicities are 1, 2. Let D be the total internal deficit below four and E the total external excess above one. The incidence sums give

$$D = (4c - 2h - 6)m + o(m), \quad (5.7)$$

$$E = (c + h - 3)m + o(m). \quad (5.8)$$

Without the congruence restriction, the available pair-count slack is

$$(5c - h - 18)m + o(m). \quad (5.9)$$

If every multiplicity is $1 \pmod{3}$, the internal values nearest four are 1, 4, so the minimum pair count rises by D ; the external values nearest one are 1, 4, so it rises by E . Feasibility would therefore require

$$5c - h - 18 \geq \frac{D + E}{m} + o(1) = 5c - h - 9 + o(1),$$

which is impossible. $\square$

Proof of Theorem 1.1. Suppose the assertion fails. For some $\varepsilon > 0$, pass to a sequence with

$$k \leq 3m^2 + (4 - 3\varepsilon)m.$$

The leading real inequality has a unique equality point at $T/m = 6$, so $T = 6m + o(m)$. The first-order form of the integer pair condition then gives $T - 6m = O(1)$; pass to a subsequence on which this difference is a fixed integer h .

The total deviation from the two balanced degree distributions is $O(m)$. Hence only $O(q)$ dual lines fail to meet $\mathcal{L}^*$ in $1 \pmod{3}$ points. Szőnyi–Weiner’s repair theorem changes $\mathcal{L}^*$ at a fixed number r of support points to obtain an exact $1 \pmod{3}$ multiset. Lemma 5.2 excludes $r = 0$. Lemma 5.1 gives

$$c \geq L(h) + r = 3 - h + r.$$

By (5.4), $r \geq 2$ when $h \equiv 1 \pmod{3}$, and $r \geq 1$ otherwise. Together with the pair-count bound $c \geq (18 + h)/5$, this yields

$$c \geq \min_{h \in \mathbb{Z}} \max \left\{ \frac{18 + h}{5}, 3 - h + r_{\min}(h) \right\} = 4, \quad (5.10)$$

where $r_{\min}(h) = 2$ for $h \equiv 1 \pmod{3}$ and is one otherwise. This contradicts the chosen sequence and proves (1.1). $\square$

6 The characteristic-two family

We turn to the ordered factorization $(u, v) = (2, 1)$. Thus

$$q = 8m, \quad n = 6m + 1, \quad \lambda = 2.$$

Near equality write

$$k = 32m^2 + cm + o(m), \quad T = 32m + h + o(1). \quad (6.1)$$

The integer pair condition and the external incidence count give

$$6c - h \geq 76, \quad e = 2(c + h - 8)m + o(m), \quad (6.2)$$

where $e = \sum_{Q \notin \mathcal{K}} (\sigma_n(Q) - 2)$ is the total excess above the required two n -secants.

The dual set $\mathcal{L}^*$ has only $O(q)$ odd-secants. By Szőnyi–Weiner [10, Theorem 1.1], it is the symmetric difference

$$\mathcal{L}^* = \mathcal{E} \triangle R, \quad (6.3)$$

where every line meets $\mathcal{E}$ in an even number of points and $|R| = r = O(1)$.

Lemma 6.1 (The even-type case). *If $R = \emptyset$ in (6.3), then $c \geq 16$.*

Proof. The balanced internal multiplicities are five and six, and the balanced external multiplicity is two. Let D be the internal deficit below six. The incidence equations and the unrestricted integer pair bound give

$$\begin{aligned} D &= (6c - 6h - 32)m + o(m), \\ e &= 2(c + h - 8)m + o(m), \\ \text{available pair slack} &= (12c - 2h - 152)m + o(m). \end{aligned}$$

If every multiplicity is even, restricting the two balanced distributions to even integers raises their minimum pair sum by

$$\frac{D + e}{2} = (4c - 2h - 24)m + o(m).$$

The available slack must be at least this large, which simplifies to $c \geq 16$. $\square$

Lemma 6.2 (Parity of the repair). *Suppose $R \neq \emptyset$ in (6.3). If r is odd, then*

$$c + h \geq 9. \quad (6.4)$$

If r is even, then

$$c + h \geq 10. \quad (6.5)$$

Moreover, $r \equiv T \equiv h \pmod{2}$.

Proof. Every even-type set has even cardinality, so $r \equiv |\mathcal{L}^*| = T \pmod{2}$; the leading term $32m$ is even, giving the last congruence.

Dualize the points of R to r lines of the original plane. On each such line, at most $r - 1$ points can cancel with the other generators in the symmetric difference. Thus at least $q - r + 2$ points have odd n -secant multiplicity. At most $n = 3q/4 + 1$ of them lie in $\mathcal{K}$, so a single generator contributes at least $q/4 - r + 1$ external points to the excess e .

If r is odd, then $r \geq 1$ and hence $e \geq q/4 - O(1) = 2m - O(1)$. If r is even, then $r \geq 2$; taking the union of two generator lines loses only their intersection, and $e \geq q/2 - O(1) = 4m - O(1)$. Comparing with (6.2) proves (6.4) and (6.5). $\square$

Proof of Theorem 1.3. Suppose the theorem fails. Pass to a sequence for which the coefficient c in (6.1) is bounded above by $73/6 - 8\varepsilon$. As in Lemma 4.2, the leading incidence bound forces $T = 32m + o(m)$, and the first-order integer condition bounds $T - 32m$. Pass to a subsequence with fixed integer difference h .

The total deviation from the balanced multiplicities is $O(m)$, so Szőnyi–Weiner’s theorem gives (6.3) with fixed r . Lemma 6.1 excludes $r = 0$ in the range under consideration. If h is odd, Lemma 6.2 and (6.2) give

$$c \geq \max \left\{ \frac{76+h}{6}, 9-h \right\}.$$

The minimum over odd h is $73/6$, attained at $h = -3$. If h is even, the corresponding bound is

$$c \geq \max \left\{ \frac{76+h}{6}, 10-h \right\},$$

whose minimum over even h is $37/3$, attained at $h = -2$. Both contradict the chosen sequence. Since $q = 8m$, this proves (1.7). $\square$

The parity argument uses only the repair theorem and the fact that an n -secant contains at most n points of the arc. It does not require a classification of even-type sets of size $4q + O(1)$.

7 Blocking sets and projective codes

The complement of a (k, n) -arc $\mathcal{K}$ in a projective plane of order q is a $(q + 1 - n)$ -fold blocking set. Completeness says more: each point of the complement lies on a line meeting it in exactly $q + 1 - n$ points. Thus the complement is a minimal multiple blocking set. Under this correspondence, n -secants of $\mathcal{K}$ are the $(q + 1 - n)$ -secants of its complement, and the hypothesis $\sigma_n(Q) \geq \lambda$ says that every point of the blocking set is on at least λ such secants. This is precisely the dual setting of Bishnoi–Mattheus–Schillewaert [2, Section 8].

In particular, Theorem 1.1 is equivalent to the following statement.

Corollary 7.1. *Let $q = 3^h$ and let B be the complement of a complete $(k, 2q/3 + 1)$ -arc in $\text{PG}(2, q)$. Then B is a minimal $q/3$ -fold blocking set and*

$$|B| \leq \frac{2q^2}{3} - \frac{q}{3} + 1 + o(q).$$

There is also a standard coding interpretation; compare Kohnert’s incidence criterion for projective code extension [5]. Projective representatives of the points of a (k, n) -arc form the columns of a generator matrix for a projective $[k, 3, k - n]_q$ code. A line meeting the arc in n points corresponds to a minimum-weight projective codeword. Adding a new projective column raises the minimum distance by one exactly when no minimum-weight hyperplane contains that column. Hence completeness is equivalent to nonextendibility. The condition $\sigma_n(Q) \geq \lambda$ records at least λ minimum-weight hyperplanes through every candidate column.

The same statement is available in every projective dimension. Points and hyperplanes of $\text{PG}(r - 1, q)$ form a symmetric design with parameters

$$v = \frac{q^r - 1}{q - 1}, \quad K = \frac{q^{r-1} - 1}{q - 1}, \quad \Lambda = \frac{q^{r-2} - 1}{q - 1}. \quad (7.1)$$

Corollary 7.2 (Projective codes). *Let C be a projective $[k, r, k - n]_q$ code, and let $\mathcal{S}$ be its minimum-weight hyperplanes. Partition the points of $\text{PG}(r - 1, q)$ into the k columns of C and the remaining candidate columns. For every integer $T = |\mathcal{S}|$, the two minimum-hyperplane degree sequences satisfy the exact interval condition (2.3), with the parameters in (7.1). Any prescribed lower bound on the number of minimum-weight hyperplanes through a candidate column may be used as the lower endpoint of its integer degree interval.*

Proof. Apply Theorem 2.2 to the point–hyperplane design of $\text{PG}(r-1, q)$. A minimum-weight hyperplane contains exactly n columns, so the selected blocks have constant intersection size n with the first point class. $\square$

Exact two-valued hyperplane intersections correspond to projective two-weight codes [3]. In the simultaneously tight families of Proposition 4.3, all but $O(q)$ dual lines have the two intersection numbers

$$uv \quad \text{and} \quad (u+1)(v+1).$$

Their difference is $u+v+1$, which must be a power of the characteristic along an infinite Desarguesian family. This agrees with the classical divisibility condition for exact projective two-weight codes. The present argument does not classify the $O(q)$ exceptional lines.

8 Concluding remarks

The first two incidence moments already contain the classical spectral bound. Their sharp integer degree ranges retain information that the real relaxation loses, and at each rational equality parameter this loss has linear order. Modular stability then converts the nearly two-valued dual intersection distribution into a bounded repair. Completeness assigns a further linear cost to every point in that repair support.

The characteristic-three result leaves a concrete extremal problem:

$$t_{2q/3+1}(2, q) \stackrel{?}{=} \frac{q^2}{3} + \frac{4q}{3} + o(q) \quad (q = 3^h \rightarrow \infty). \quad (8.1)$$

A matching construction would determine the first two asymptotic terms. A second problem is structural: classify the sets of dual points with only $O(q)$ lines outside the two predicted intersection numbers and determine which of them can occur as the full set of n -secants of a complete (k, n) -arc.

AI assistance disclosure

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